

In 2020, how did higher percentages of frequent physical or mental distress moderate the relationship between environmental/ socioeconomic factors and COVID19 cases?

```
In [24]: #! pip install -q pandas  
#! pip install -q pyppeteer  
#! pip install -q qeds  
##pip install numpy  
#! pip install -q pandas  
#! pip install -q pyppeteer  
#! pip install -q qeds  
#! pip install -q matplotlib  
#! pip install -q tabulate  
from tabulate import tabulate  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import datetime as dt  
import qeds
```

Introduction

- Source of Data
 - My dataset comes from a combination of data sources. The socioeconomic factors data comes from the US CDC metrics of "social vulnerability" and natural and man-made disasters. The weather data comes from the NOAA Global Surface Summary of the day, which links each county to a nearby weather station. Also, my dataset uses the County Health Rankings Data a dataset that contains many measures of community health.
- Background
 - Many studies have studied the link between weather and COVID19 cases. The consensus is usually that warmer weather leads to more cases, but the issue is that socioeconomic status could be different in warmer areas and that could be a confounder. In my research, I want to affirm that relationship plus explore an additional aspect of consideration known as physical and mental stress. I want to see how this factor interacts with what we already know.
- Research Question
 - In 2020, how did higher percentages of frequent physical or mental distress moderate the relationship between environmental/socioeconomic factors and COVID19 cases across US counties, specifically California?

- Findings
 - (put this part later)
- Citations
 - Benita, F., Rebollar-Ruelas, L., & Gaytán-Alfaro, E. D. (2022). What have we learned about socioeconomic inequalities in the spread of COVID-19? A systematic review. *Sustainable cities and society*, 86, 104158. <https://doi.org/10.1016/j.scs.2022.104158>
 - Han, Y., Zhao, W., & Pereira, P. (2022). Global COVID-19 pandemic trends and their relationship with meteorological variables, air pollutants and socioeconomic aspects. *Environmental research*, 204(Pt C), 112249. <https://doi.org/10.1016/j.envres.2021.112249>
 - Sarkodie, S. A., & Owusu, P. A. (2021). Global effect of city-to-city air pollution, health conditions, climatic & socio-economic factors on COVID-19 pandemic. *The Science of the total environment*, 778, 146394. <https://doi.org/10.1016/j.scitotenv.2021.146394>
 - Ahmed, J., Jaman, M. H., Saha, G., & Ghosh, P. (2021). Effect of environmental and socio-economic factors on the spreading of COVID-19 at 70 cities/provinces. *Heliyon*, 7(5), e06979. <https://doi.org/10.1016/j.heliyon.2021.e06979>
 - Zhang, X., Maggioni, V., Houser, P., Xue, Y., & Mei, Y. (2022). The impact of weather condition and social activity on COVID-19 transmission in the United States. *Journal of environmental management*, 302(Pt B), 114085. <https://doi.org/10.1016/j.jenvman.2021.114085>
 - Ganslmeier, M., Furceri, D., & Ostry, J. D. (2021). The impact of weather on COVID-19 pandemic. *Scientific reports*, 11(1), 22027. <https://doi.org/10.1038/s41598-021-01189-3>
- Variables
 - X-variables = mean_temp, dewpoint, windspeed, percent_frequent_physical_distress, percent_frequent_mental_distress, percent_no_highschool_diploma, average_daily_pm2_5, unemployment_rate, high_school_graduation_rate, percent_limited_access_to_healthy_foods, personal income, real_gdp_2020
 - Y-variable = Cases
- Rationale
 - I chose these variables because of their possible connection to COVID19 cases. Some studies like Ganslmeier et al. (2021) suggest a negative correlation between temperature, humidity, windspeed to COVID19 cases and I wish to explore that correlation with my temperature, dewpoint, and wind speed variables. Also, since Han et al. (2020) mention that the interaction of different types of pollution has a strong impact on cases, I included the PM2.5 variable to explore that effect. The physical and mental distress variables are there to add an additional vector for exploration. I wonder how those variables can interact with the already explored effects of other variables on COVID19 cases. Then I included some socioeconomic variables to cover their potential effect on COVID19 cases. For example, Benita et al.

(2022) concludes that income inequality is positively correlated with transmission of the virus. I included education variables, measures of personal income, GDP, unemployment rate as socioeconomic variables. I think it is valuable to explore this since many research have stated there are limitations reducing the accuracy of their research.

Data Cleaning and Loading

```
In [25]: # Load datasets and do some filtering

# Load main data and filter for California only
covid_health_weather = pd.read_csv("c:/Users/Pikachu/OneDrive - University of Toron
covid_health_weather = covid_health_weather[covid_health_weather['state'] == 'Calif

# Get the number of rows and columns in the dataframe
num_rows, num_columns = covid_health_weather.shape
print(f'The dataframe has {num_rows} rows and {num_columns} columns.')

# Load the data for California GDP and Income and filter for 2020 only
# Remove all year columns except for 2020
ca_gdp_2020 = pd.read_csv("c:/Users/Pikachu/OneDrive - University of Toronto/W 2024
ca_gdp_2020 = ca_gdp_2020[['GeoFIPS', 'GeoName', 'Region', 'TableName', 'LineCode',

ca_inc_2020 = pd.read_csv("c:/Users/Pikachu/OneDrive - University of Toronto/W 2024
ca_inc_2020 = ca_inc_2020[['GeoFIPS', 'GeoName', 'Region', 'TableName', 'LineCode',

ca_unemp_2020 = pd.read_csv("c:/Users/Pikachu/OneDrive - University of Toronto/W 20
ca_unemp_2020 = ca_unemp_2020[['FIPS', 'Name', '2020']]

ca_gdp_2020 = ca_gdp_2020[ca_gdp_2020['LineCode'] == 1]
ca_pop_2020 = ca_inc_2020[ca_inc_2020['LineCode'] == 2]
ca_inc_2020 = ca_inc_2020[ca_inc_2020['LineCode'] == 3]
# Rename the 2020 column to real_gdp_2020
ca_gdp_2020.rename(columns = {'2020': 'real_gdp_2020'}, inplace = True)
# Rename to 2020 column to personal_income_capita_2020
ca_inc_2020.rename(columns = {'2020': 'personal_income_capita_2020'}, inplace = Tru
# Rename the 2020 column to unemployment_rate_2020
ca_unemp_2020.rename(columns = {'2020': 'unemployment_rate_2020'}, inplace = True)

ca_pop_2020.rename(columns = {'2020': 'population_2020'}, inplace = True)

# Convert 'FIPS' column to string and remove any extra characters
ca_unemp_2020['FIPS'] = ca_unemp_2020['FIPS'].astype(str).str.replace("'", '').str.
ca_gdp_2020['GeoFIPS'] = ca_gdp_2020['GeoFIPS'].astype(str).str.replace("'", '').st
ca_inc_2020['GeoFIPS'] = ca_inc_2020['GeoFIPS'].astype(str).str.replace("'", '').st
ca_pop_2020['GeoFIPS'] = ca_pop_2020['GeoFIPS'].astype(str).str.replace("'", '').st

# Convert 'fips' column to string to ensure the keys are of the same type
covid_health_weather['fips'] = covid_health_weather['fips'].astype(str).str.replace

# Merge the new datasets into the main dataset using inner join
```



```

+-----+-----+-----+-----+
--+
|          |          |          | count |          |          |          | st
d          | min      | 25%      | 50%    |          | 75%      | max
|
+-----+-----+-----+-----+
--+
|          |          |          | 15226.0 | 8284.8495993695 | 29036.546
069279182 | 1.0      | 57.0      | 621.0    | 5397.75      | 430713.0
|
|          |          |          | 14250.0 | 66.0297894736842 | 12.872566
688379507 | 14.5     | 57.2      | 65.85    | 74.6         | 109.6
|
|          |          |          | 15227.0 | 57187227.536875285 | 12412368
3.98597665 | 91114.0  | 2026350.0 | 9115222.0 | 44416463.0 | 2933320241.
0 |
|          |          |          | 15227.0 | 63159.115321468445 | 24574.379
249841168 | 33262.0  | 47615.0   | 56163.0  | 68907.0     | 145393.0
|
|          |          |          | 13213.0 | 44.34217815787482 | 10.670153
282176337 | -9.3     | 38.3      | 46.3     | 52.4        | 76.1
|
|          |          |          | 13740.0 | 5.268901018922853 | 2.8238938
88671211  | 0.0      | 3.4       | 4.7      | 6.5         | 25.9
|
|          |          |          | 15226.0 | 11.197254722593557 | 1.5431733
118128534 | 8.3447137219 | 10.09799344 | 11.063575791 | 12.424381592 | 15.303700776
|
|          |          |          | 15226.0 | 12.074587089343472 | 1.3516725
846252653 | 8.7135832655 | 11.265630833 | 11.969444594 | 13.254879616 | 15.037081267
|
|          |          |          | 15226.0 | 16.435820307368974 | 7.1973315
78157117  | 5.8      | 11.0      | 13.7     | 21.2        | 33.0
|
|          |          |          | 15226.0 | 10.239570471561803 | 2.8543892
831960784 | 5.6      | 8.7       | 9.6      | 11.0        | 19.7
|
|          |          |          | 14976.0 | 82.47739001227511 | 10.493180
50918275  | 36.301369863 | 81.374853114 | 85.239238869 | 87.432808156 | 100.0
|
|          |          |          | 15227.0 | 9.898778485584817 | 2.4663815
693111415 | 6.8      | 8.3       | 9.4      | 11.1        | 22.6
|
|          |          |          | 15226.0 | 5.9155351743400955 | 4.376067
38752614  | 0.3219526811 | 3.0248215663 | 4.7694055595 | 8.004312443 | 27.847524081
|
+-----+-----+-----+-----+
--+

```

- Mean_temp is mean temperature. The median appears to be around 18.3 degrees celsius or 65 degrees fahrenheit, but temperature can vary between 14.5 F and 109.6 F. We will connect these varying temps to cases through analysis

- cases counts number of covid19 cases for that city on that day. This will be our main response variable
- real gdp in 2020 is expressed in terms of 2017 dollars. It is one of the key statistics in measuring economic wellbeing of a county and will be one our socioeconomic variables that we are connecting to covid19 cases. GDP could be negatively correlated with cases
- We use personal income to measure income as a socioeconomic variable that can connect to number of cases. Lower income could lead to more cases
- Dewpoint is the temperature the air needs to be cooled to reach a relative humidity of 100%. This is a measure of the moisture in the air. We will use this to predict cases
- Percent frequent and mental distress potentially interacts with cases. We will explore how higher levels of distress impact cases. We will explore how this number interacts with the other factors
- High school graduation rate, unemployment rate, access to healthy goods are some rate based economic variables we will explore on how they are correlated with cases

```
In [27]: # Define thresholds for high physical and mental distress
high_physical_distress_threshold = covid_health_weather['percent_frequent_physical_
high_mental_distress_threshold = covid_health_weather['percent_frequent_mental_dist

# Filter the data for high physical and mental distress
high_distress_data = covid_health_weather[
    (covid_health_weather['percent_frequent_physical_distress'] >= high_physical_di
    (covid_health_weather['percent_frequent_mental_distress'] >= high_mental_distre
]

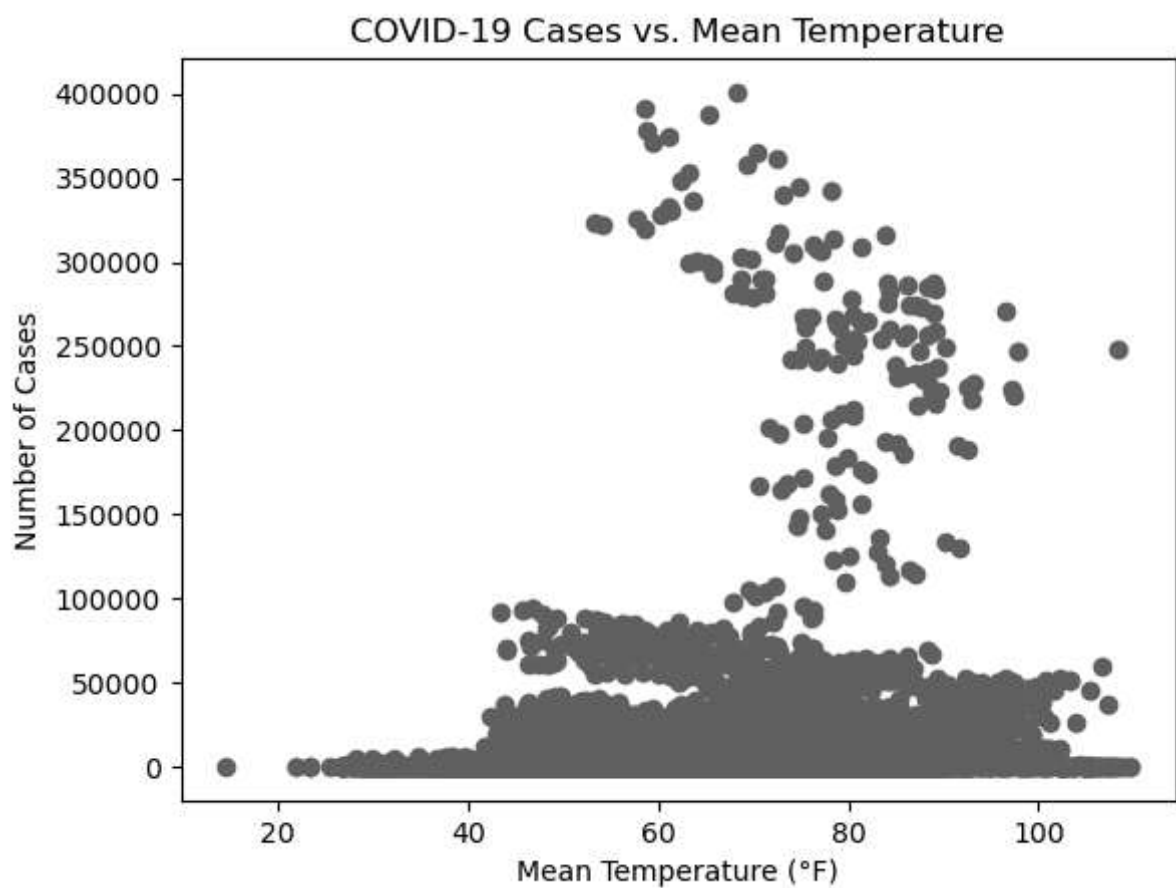
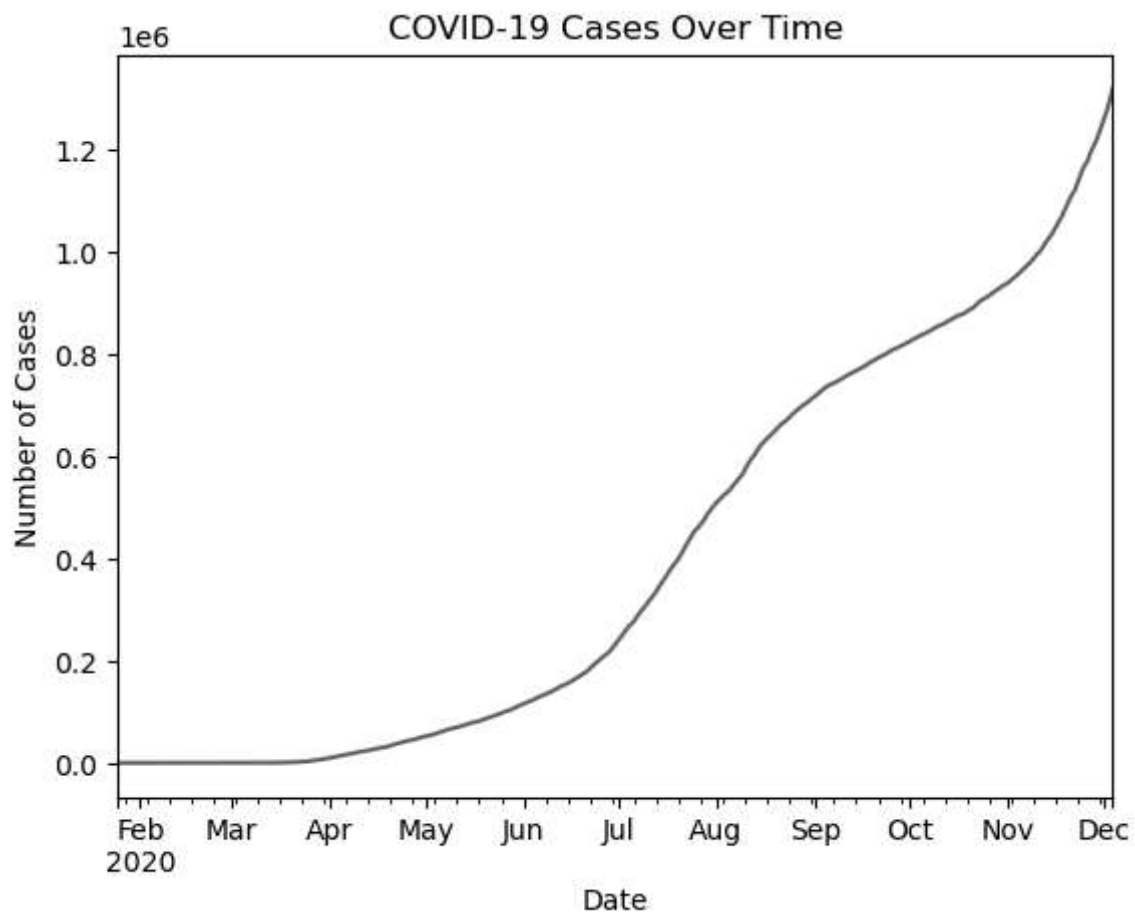
summary_stats_hd = high_distress_data[['cases', 'mean_temp', 'real_gdp_2020', 'pers
    'percent_frequent_physical_distress', 'perce
    'average_daily_pm2_5', 'high_school_gradua
    'percent_limited_access_to_healthy_foods
summary_stats_hd = summary_stats_hd.transpose()
formatted_table_hd = tabulate(summary_stats_hd, headers='keys', tablefmt='pretty')
print(formatted_table_hd)
```

	min	25%	50%	count	mean	std
				75%	max	
cases				2774.0	4048.8702235039655	7687.30293
8987098	1.0	22.0	339.0	4296.0	43860.0	
mean_temp				2608.0	67.95548312883436	13.8746639
25216002	14.5	56.8	68.6	78.9	102.3	
real_gdp_2020				2774.0	9628559.453136265	12763540.9
61196307	107899.0	783182.0	6201866.0	9115222.0	45610265.0	
personal_income_capita_2020				2774.0	46779.38536409517	9428.51198
6266405	33262.0	43949.0	44409.0	45909.0	74115.0	
dewpoint				2118.0	44.64981114258734	10.1027890
31203308	-3.9	38.425	45.9	51.775	76.1	
wind_speed				2357.0	4.9843869325413666	2.5466201
0427211	0.2	3.3	4.6	6.0	18.1	
percent_frequent_physical_distress				2774.0	13.299212080714854	0.82617112
70894292	12.543376138	12.659281899	12.818467125	13.740965619	15.303700776	
percent_frequent_mental_distress				2774.0	13.775850397510094	0.47421012
45105653	13.265377808	13.477388425	13.612969235	13.912127973	15.037081267	
percent_no_highschool_diploma				2774.0	21.47548666186013	8.85086586
5845599	8.9	11.7	17.8	31.4	33.0	
average_daily_pm2_5				2774.0	11.39408795962509	4.40094091
5137151	5.6	7.8	8.9	14.0	19.7	
high_school_graduation_rate				2524.0	83.66429266733638	4.04616846
8430391	74.358974359	80.97826087	85.245901639	86.146559663	88.482023968	
unemployment_rate_2020				2774.0	11.951405912040373	3.77555910
59055735	8.2	9.4	11.1	12.8	22.6	
percent_limited_access_to_healthy_foods				2774.0	9.019512184043618	2.32963612
76862254	5.3756534862	7.4108434196	8.8417960189	9.8929138731	14.001309659	

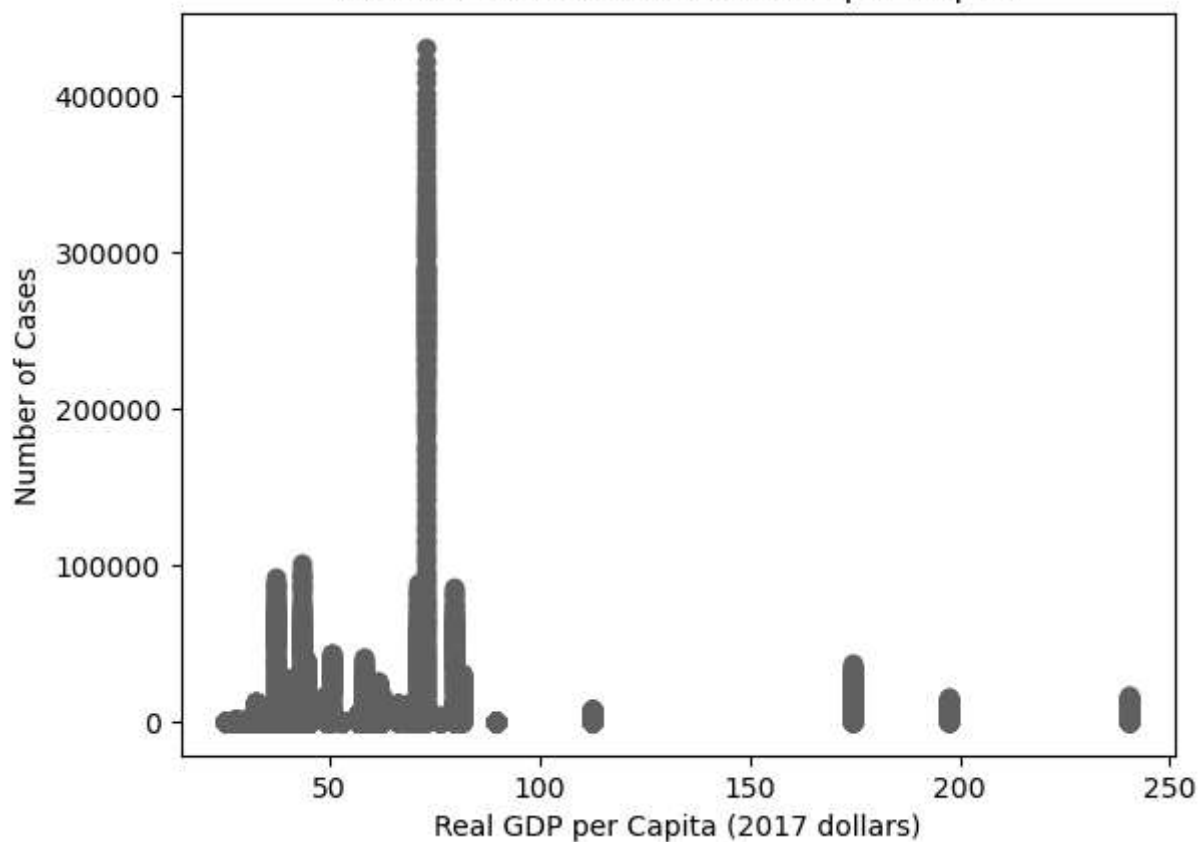
- This table is the summary statistics for the subgroup where distress is higher than the 75th quantile. I will use this subgroup to explore the effect of high distress on cases

Plots

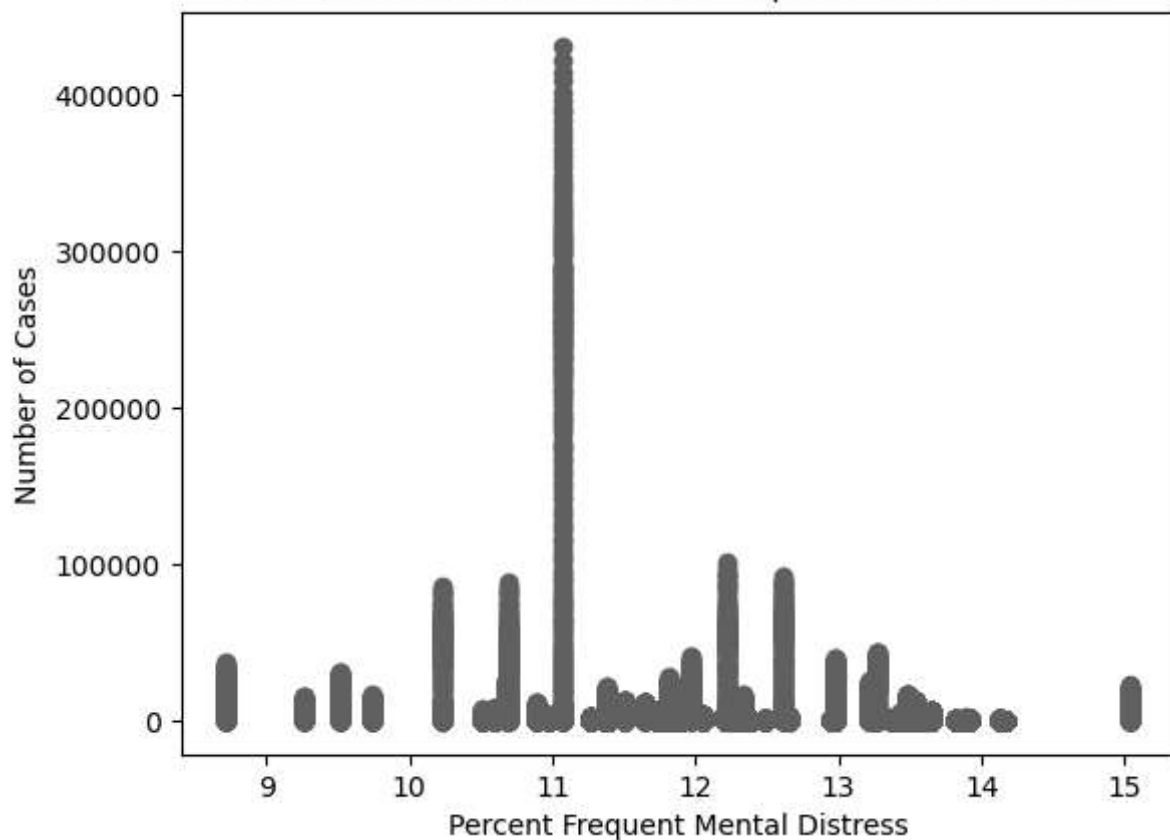
```
In [28]: # 1. **COVID-19 Cases Over Time**:  
#       - Plot the number of COVID-19 cases over time to see the trend.  
  
covid_health_weather.groupby('date')['cases'].sum().plot(kind='line', title='COVID-  
plt.xlabel('Date')  
plt.ylabel('Number of Cases')  
plt.show()  
  
# 2. **COVID-19 Cases vs. Mean Temperature**:  
#       - Scatter plot to see the relationship between COVID-19 cases and mean temper  
  
plt.scatter(covid_health_weather['mean_temp'], covid_health_weather['cases'])  
plt.title('COVID-19 Cases vs. Mean Temperature')  
plt.xlabel('Mean Temperature (°F)')  
plt.ylabel('Number of Cases')  
plt.show()  
  
# 3. **COVID-19 Cases vs. Real GDP per Capita**:  
#       - Scatter plot to see the relationship between COVID-19 cases and real GDP pe  
  
plt.scatter(covid_health_weather['real_gdp_per_capita_2020'], covid_health_weather[  
plt.title('COVID-19 Cases vs. Real GDP per Capita')  
plt.xlabel('Real GDP per Capita (2017 dollars)')  
plt.ylabel('Number of Cases')  
plt.show()  
  
# 5. **COVID-19 Cases vs. Percent Frequent Mental Distress**:  
#       - Scatter plot to see the relationship between COVID-19 cases and percent fre  
  
plt.scatter(covid_health_weather['percent_frequent_mental_distress'], covid_health_  
plt.title('COVID-19 Cases vs. Percent Frequent Mental Distress')  
plt.xlabel('Percent Frequent Mental Distress')  
plt.ylabel('Number of Cases')  
plt.show()  
  
# 6. **COVID-19 Cases vs. Unemployment Rate**:  
#       - Scatter plot to see the relationship between COVID-19 cases and unemploymen  
  
plt.scatter(covid_health_weather['unemployment_rate_2020'], covid_health_weather['c  
plt.title('COVID-19 Cases vs. Unemployment Rate')  
plt.xlabel('Unemployment Rate (%)')  
plt.ylabel('Number of Cases')  
plt.show()
```



COVID-19 Cases vs. Real GDP per Capita



COVID-19 Cases vs. Percent Frequent Mental Distress



COVID-19 Cases vs. Unemployment Rate

